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A Model details

In our experiment, we tested 20 modern LLM/LRMs. All experiments with temperature set to zero.

- **Deepseek-V3 (671B)** (DeepSeek-AI et al., 2024) is a state-of-the-art open-source LLM released by Deepseek.
- **Deepseek-R1 (671B)** (DeepSeek-AI et al., 2025) is a leading open-source LRM trained with reinforcement learning using a rule-based reward system.
- **Gemma-2-9B / Gemma-2-27B** (Gemma Team et al., 2024) is an open-source, lightweight yet high-performance LLM series.
- **Llama-3.1-8B / Llama-3.1-70B / Llama-3.1-405B** (Meta, 2024) is an open-source dense model series incorporating Direct Preference Optimization (DPO) (Rafailov et al., 2024) for alignment.
- **Mistral-7B Instruct v0.3** (Jiang et al., 2023) is an early high-performance and lightweight open-source LLM.
- **Mistral Small 3 (24B)** (Mistral AI, 2025) is the latest high-performance open-source LLM from Mistral, designed for efficiency.
- **Qwen-2.5-7B / Qwen-2.5-32B / Qwen-2.5-72B** (Qwen et al., 2025) is an open-source MoE LLM series pre-trained on 18 trillion tokens and fine-tuned with 1 million examples.
- **QwQ-32B** (Qwen Team, 2025) is a reasoning-focused LLM trained via reinforcement learning, achieving competitive performance through scalable RL and agent-integrated reasoning for tool use and environmental adaptation.
- **Gemini-1.5-flash / Gemini-1.5-pro** (Google, 2024) is a proprietary MoE LLM series optimized for processing ultra-long sequences.
- **Gemini-2.0-flash** (DeepMind, 2024) is the latest proprietary LLM in the Gemini series, featuring enhanced multimodal understanding and reasoning capabilities.
- **GPT-3.5-turbo** (OpenAI, 2022) is a proprietary conversational LLM fine-tuned via RLHF (Ouyang et al., 2022) and PPO (Schulman et al., 2017).
- **GPT-4o-mini / GPT-4o** (OpenAI, 2024) is a proprietary multimodal LLM from OpenAI with enhanced reasoning capabilities.
- **o1-mini** (OpenAI, 2024) is a proprietary LRM from OpenAI utilizing reinforcement learning for inference-time scaling.